

Safety assessment of the process Starlinger viscotec deCON used to recycle post-consumer PET into food contact materials

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The declarations of interest of all scientific experts active in EFSA's work are available at <https://open.efsa.europa.eu/experts>.

Abstract

The EFSA Panel on Food Contact Materials (FCM) assessed the safety of the recycling process Starlinger viscotec deCON (EU register number RECYC339). The input is washed and dried poly(ethylene terephthalate) (PET) flakes originating from collected post-consumer PET containers, with no more than 5% PET from non-food consumer applications. The flakes are preheated batchwise before being submitted to solid state polycondensation (SSP) in batch or semi-continuous reactor(s), at high temperature under vacuum and gas flow. Having examined the challenge test provided, the Panel concluded that the preheating (step 2) and the decontamination in the SSP reactor (step 3) are critical in determining the decontamination efficiency of the process. The operating parameters to control the performance are temperature, residence time, pressure and gas flow rate. It was demonstrated that this process ensures that the level of migration of potential unknown contaminants into food is below the conservatively modelled migration of 0.0481 or 0.0962 µg/kg food (depending on the molar mass of the contaminant substance) derived from the exposure scenario for infants when such recycled PET is used at up to 55% together with virgin PET, and of 0.156 or 0.312 µg/kg food, derived from the exposure scenario for toddlers, when used at up to 100%. The Panel concluded that the recycled PET obtained from this process is not of safety concern, when used at up to 55% in mixture with virgin PET for manufacturing materials and articles for contact with all types of foodstuffs, including drinking water, and at up to 100% for contact with all types of foodstuffs except drinking water, and used for long-term storage at room temperature or below, with or without hot-fill. Articles made of this recycled PET are not intended to be used in microwave and conventional ovens and such uses are not covered by this evaluation.

KEYWORDS

food contact materials, plastic, poly(ethylene terephthalate) (PET), recycling process, safety assessment, Starlinger & Co. GmbH, Starlinger viscotec deCON

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1 | INTRODUCTION

1.1 | Background

Recycled plastic materials and articles shall only be placed on the market if the recycled plastic is from an authorised recycling process. Before a recycling process is authorised, the European Food Safety Authority (EFSA)'s opinion on its safety is required. This procedure has been established in Articles 17 and 18 of Commission Regulation (EU) 2022/1616¹ on recycled plastic materials intended to come into contact with foods. More specifically, according to Article 18 of Commission Regulation (EU) 2022/1616 on recycled plastic materials intended to come into contact with foods, EFSA is required to carry out risk assessments on the risks originating from the migration of substances from recycled food contact plastic materials and articles into food, to evaluate the microbiological safety of these materials and articles and to deliver a scientific opinion on the recycling process examined.

According to this procedure, the process developers submit applications to the competent authorities of Member States, which transmit the applications to EFSA for evaluation. In this case, EFSA received an application from the Austrian Competent Authority (Bundesministerium für Soziales, Gesundheit, Pflege und Konsumentenschutz), for evaluating the recycling process Starlinger viscotec deCON, European Union (EU) register No RECYC339. The request has been registered in the EFSA's register of received questions under the number EFSA-Q-2023-00462. The dossier was submitted on behalf of Starlinger & Co. GmbH, Division viscotec, Adsdorf 384113 St. Martin im Mühlkreis, Austria (see '[Documentation provided to EFSA](#)').

1.2 | Terms of Reference

The Austrian Competent Authority (Bundesministerium für Soziales, Gesundheit, Pflege und Konsumentenschutz), requested the safety evaluation of the recycling process Starlinger viscotec deCON, in compliance with Article 17 of Commission Regulation (EU) 2022/1616. The recycling process uses the recycling technology number 1 of the list of suitable recycling technologies of Table 1 of Annex 1 of Commission Regulation (EU) 2022/1616.

2 | DATA AND METHODOLOGIES

2.1 | Data

The applicant submitted a confidential and a non-confidential version of a dossier, following EFSA's 'Scientific Guidance on the criteria for the evaluation and on the preparation of applications for the safety assessment of post-consumer mechanical PET recycling processes intended to be used for manufacture of materials and articles in contact with food' (EFSA CEP Panel, 2024) and EFSA's 'Administrative guidance for the preparation of applications for the authorisation of individual recycling processes to produce recycled plastics materials and articles intended to come into contact with food' (EFSA, 2024).

Additional information was received from the applicant during the assessment process, in response to requests from EFSA sent on 6 February 2024, 27 November 2024 and 20 May 2025 (see '[Documentation provided to EFSA](#)'). A technical hearing was held with the applicant on 13 May 2025.²

In accordance with Art. 38 of the Regulation (EC) No 178/2002³ and taking into account the protection of confidential information and the personal data in accordance with Articles 39 to 39e of the same Regulation, and of the Decision of the EFSA's Executive Director laying down practical arrangements concerning transparency and confidentiality,⁴ the non-confidential version of the dossier has been published on Open.EFSA.⁵

According to Art. 32c(2) of Regulation (EC) No 178/2002 and to the Decision of EFSA's Executive Director laying down the practical arrangements on pre-submission phase and public consultations,⁴ EFSA carried out a public consultation on the non-confidential version of the application from 15 July to 05 August 2024 for which no comments were received.

The following information on the recycling process was provided by the applicant and used for the evaluation (EFSA, 2024; EFSA CEP Panel, 2024):

¹Commission Regulation (EU) 2022/1616 of 15 September 2022 on recycled plastic materials and articles intended to come into contact with foods, and repealing Commission Regulation (EC) No 282/2008. OJ L 243, 20.9.2022, p. 3–46.

²Minutes of the 75th Working Group meeting on Recycling Plastics. <https://www.efsa.europa.eu/sites/default/files/wgs/food-ingredients-and-packaging/recycling-plastics-min.pdf>.

³Commission Regulation (EC) No 178/2002 of the European Parliament and of the Council of 28 January 2002 laying down the general principles and requirements of food law, establishing the European Food Safety Authority and laying down procedures in matters of food safety. OJ L 31, 1.2.2002, p. 1–48.

⁴Decision available at: <https://www.efsa.europa.eu/en/corporate-pubs/transparency-regulation-practical-arrangements>.

⁵The non-confidential version of the dossier has been published on Open.EFSA and is available at the following link: <https://open.efsa.europa.eu/dossier/FCM-2023-17478>.

- Recycling process,
- Determination of the decontamination efficiency of the recycling process,
- Table of operating parameters,
- Self-evaluation of the recycling process.

2.2 | Methodologies

The risks associated with the use of recycled plastic materials and articles in contact with food come from the possible migration of chemicals into the food in amounts that would endanger human health. The quality of the input, the efficiency of the recycling process to remove contaminants as well as the intended use of the recycled plastic are crucial points for the risk assessment (EFSA CEP Panel, 2024).

The criteria for the safety evaluation of a mechanical recycling process to produce recycled PET intended to be used for the manufacture of materials and articles in contact with food are described in the scientific guidance developed by the EFSA Panel on Food Contact Materials, Enzymes and Processing Aids (EFSA CEP Panel, 2024). The principle of the evaluation is to apply the decontamination efficiency of a recycling process, obtained from a challenge test with surrogate contaminants, to a reference contamination level for post-consumer PET, conservatively set at 3 mg/kg PET for contaminants resulting from possible misuse. The resulting residual concentration of each surrogate contaminant in recycled PET (C_{res}) is compared with a modelled concentration of the surrogate contaminants in PET (C_{mod}). This C_{mod} is calculated using generally recognised conservative migration models so that the related migration does not give rise to a dietary exposure exceeding 0.0025 µg/kg body weight (bw) per day (i.e. the human exposure threshold value for chemicals with structural alerts for genotoxicity), below which the risk to human health would be negligible, considering different dietary exposure scenarios (EFSA CEP Panel, 2024). If the C_{res} is not higher than the C_{mod} , the recycled PET manufactured by such recycling process is not considered of safety concern for the defined conditions of use (EFSA CEP Panel, 2024).

The assessment was conducted in line with the principles described in the EFSA Guidance on transparency in the scientific aspects of risk assessment, considering the relevant guidance from the EFSA Scientific Committee (EFSA, 2009).

3 | ASSESSMENT

3.1 | General information⁶

According to the applicant, the recycling process Starlinger viscotec deCON (configurations named as listed in Appendix C) is intended to recycle food grade PET containers. The recycled PET is intended to be used at up to 55% in mixtures with virgin PET for the manufacture of bottles for mineral water and at up to 100% for other materials and articles for direct contact with all kinds of foodstuffs, such as sheets, thermoformed trays or cups, woven fibre and bottles for soft drinks, juices and beer. It is intended for long-term storage at room temperature or below, with or without hot-fill. The final articles are not intended to be used in microwave or conventional ovens.

3.2 | Description of the process

3.2.1 | General description⁷

The recycling process Starlinger viscotec deCON produces recycled PET flakes from PET materials originating from post-consumer collection systems.⁸

Input

- In step 1, the post-consumer PET is processed into hot washed and dried flakes.

Decontamination and production of recycled PET material

- In step 2, the flakes are preheated in batch reactor(s) by a flow of hot gas.
- In step 3, the preheated flakes are submitted to one semi-continuous or four batch solid state polycondensation (SSP) reactors running at high temperature and using a combination of vacuum and gas flow.

The operating conditions of the process have been provided to EFSA.

⁶Technical dossier, sections 'Recycling process' and 'Intended application in contact with food'.

⁷Technical dossier, sections 'Recycling process', 'Characterisation of the input' and 'Characterisation of the recycled plastic'.

⁸The collected plastic waste should comply with Art. 6 of Reg. (EC) 2022/1616.

Flakes, the final product of the process, are checked against technical requirements, such as intrinsic viscosity and the content of poly(vinyl chloride) (PVC), other plastics, glues, metals and cellulose.

3.2.2 | Characterisation of the pre-processed plastic input⁹

According to the applicant, the input material consists of hot washed and dried flakes obtained from PET materials, e.g. bottles, previously used for food packaging, from post-consumer collection systems. A small fraction may originate from non-food applications. According to the applicant, the proportion will be no more than 5%, as specified in Article 7 and Table 1 of Annex I of Commission Regulation (EU) 2022/1616.

Technical specifications on the hot washed and dried flakes were provided, such as on physical properties and residual contents of moisture, PVC, polyamide (PA), polyolefins, polycarbonate (PC), polystyrene (PS), glue, metals, wood and paper (see [Appendix A](#)).

3.3 | Starlinger viscotec deCON process

3.3.1 | Description of the main steps¹⁰

The process flow diagrams for two configurations of the Starlinger viscotec deCON process, as provided by the applicant, are reported in [Figures 1](#) and [2](#). The steps are:

- **Preheating (step 2):** The flakes are preheated in one or more batch reactors by a flow of hot gas up to the temperature of the SSP reactor.
- **SSP (step 3):** The preheated flakes are fed into the SSP reactor(s) semi-continuously or batchwise and treated under vacuum, while a gas flow is applied to support the removal of the contaminants from the flakes.

The process runs under two types of configuration:

- In the Starlinger viscotec deCON 50 type of configurations (as listed in [Appendix C; Figure 1](#)), two pre-heaters feed preheated flakes into an SSP reactor. The process is semi-continuous, since feeding the reactor occur in batches.
- In the Starlinger viscotec deCON 20 configuration ([Figure 2](#)), one pre-heater feeds four SSP reactors in sequence working in batch mode.

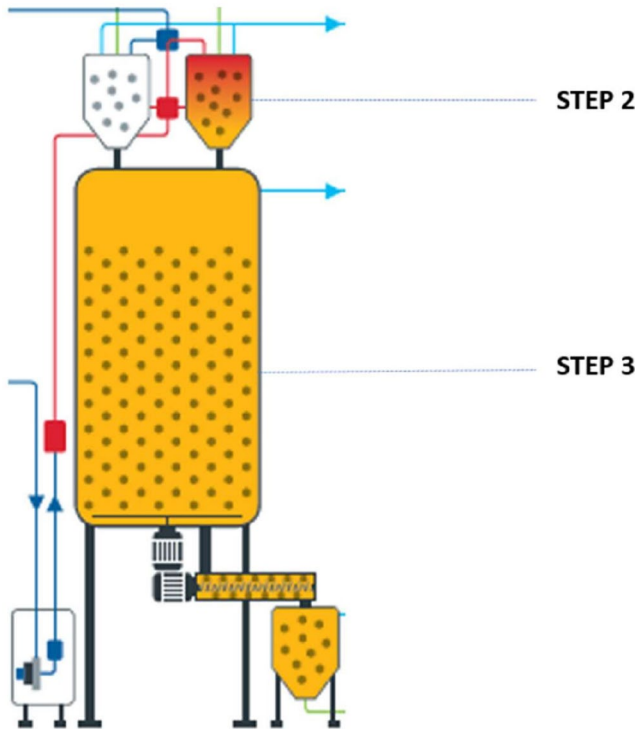


FIGURE 1 Process flow diagram of the viscotec deCON 50 configuration (provided by the applicant).

⁹Technical dossier, section 'Characterisation of the input'.

¹⁰Technical dossier, Section 'Recycling process'.

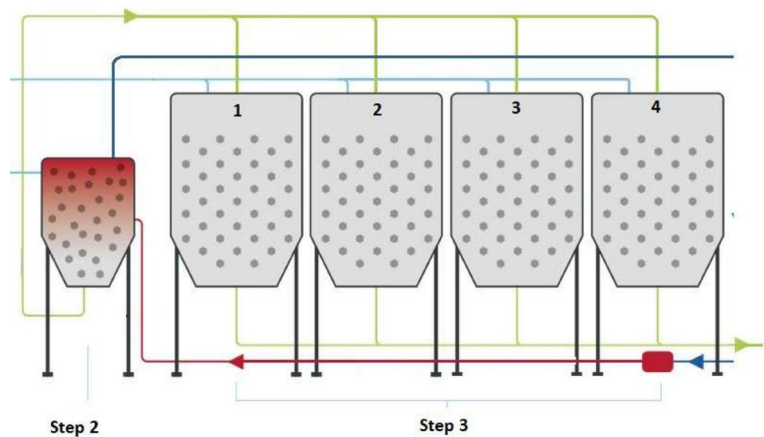


FIGURE 2 Process flow diagram of the viscotec deCON 20 configuration (provided by the applicant).

The process is run under defined operating parameters¹¹ of temperature, pressure, gas flow rate and residence time. All critical parameters are monitored [REDACTED]. The parameter values are automatically reported via the relevant software; deviations from the acceptable parameter values or ranges are alerted.¹²

3.3.2 | Decontamination efficiency of the recycling process¹³

To demonstrate the decontamination efficiency of the recycling process Starlinger viscotec deCON, a challenge test performed at industrial scale on a deCON 20 installation was submitted to EFSA.

PET flakes were contaminated with toluene, chlorobenzene, methyl salicylate, phenylcyclohexane, benzophenone and methyl stearate, selected as surrogates in agreement with the EFSA Scientific Guidance (EFSA CEP Panel, 2024) and in accordance with the recommendations of the US Food and Drug Administration (FDA, 2021).

[REDACTED] Then they were washed [REDACTED]. The concentration of the surrogates in these flakes was determined.

The washed and dried contaminated flakes (step 2) were preheated and then fed into the SSP reactor (step 3). The flakes were analysed after step 3 for their residual concentrations of the applied surrogates.

The decontamination efficiency of the process was calculated from the concentrations of the surrogates measured in the washed contaminated flakes before the preheating (before step 2) and after SSP (step 3). The results are summarised in Table 1.

TABLE 1 Efficiency of the decontamination of the Starlinger viscotec deCON process in the challenge test.

Surrogates	Concentration of surrogates before step 2 (mg/kg PET)	Concentration of surrogates after step 3 (mg/kg PET)	Decontamination efficiency (%)
Toluene	[REDACTED]	[REDACTED]	97.6
Chlorobenzene	[REDACTED]	[REDACTED]	98.0
Methyl salicylate	[REDACTED]	[REDACTED]	98.0
Phenylcyclohexane	[REDACTED]	[REDACTED]	95.2
Benzophenone	[REDACTED]	[REDACTED]	94.4
Methyl stearate	[REDACTED]	[REDACTED]	96.5

Abbreviation: PET, poly(ethylene terephthalate).

¹¹In accordance with Art. 9 and 20 of Regulation (EC) No 1935/2004 the parameters were provided to EFSA by the applicant and made available to the Member States and the European Commission (see Appendix C).
¹²Technical dossier, Sections ‘Recycling process’ and ‘Quality Assurance System’.
¹³Technical dossier, Section ‘Determination of the decontamination efficiency of the recycling process’.
¹⁴Conventional recycling commonly includes sorting, grinding, washing and drying steps and produces washed and dried flakes.

3.4 | Discussion

Considering the high temperatures used during the process, the possibility of contamination by microorganisms can be discounted. Therefore, this evaluation focuses on the chemical safety of the final product.

Specifications on the input material (i.e. washed and dried flakes, step 1) are listed in [Appendix A](#).

The flakes are produced from PET containers, e.g. bottles, previously used for food packaging, collected through post-consumer collection systems. However, a small fraction may originate from non-food applications, such as bottles for soap, mouthwash or kitchen hygiene agents. According to the applicant, the collection system and the sorting are managed in such a way that this fraction will be no more than 5% in the input,¹⁵ as recommended by the EFSA CEP Panel in its Guidance (EFSA CEP Panel, 2024).

The process is adequately described. It comprises the batch preheating (step 2) and the semi-continuous or batch SSP (step 3). The operating parameters of temperature, residence time, pressure and gas flow rate for both steps have been provided to EFSA.

A challenge test to measure the decontamination efficiency was conducted at industrial scale on process steps 2 and 3. The Panel considered that it was performed correctly according to the recommendations of the EFSA Guidance (EFSA CEP Panel, 2024). The Panel considered that steps 2 and 3 were critical for the decontamination efficiency of the process. Consequently, temperature, residence time, pressure and reduced gas flow rate of steps 2 and 3 should be controlled to guarantee the performance of the decontamination ([Appendix C](#)).

The Panel noted that in both the deCON 50 and the deCON 20 configurations there is no mixing in the SSP, since the reactors run in batches. Also, temperature, residence time, pressure and reduced gas flow rate of steps 2 and 3 are the same. Thus, it is considered that the challenge test, run in the deCON 20 equipment, is representative of the two configurations and for all sizes of the installations listed in [Appendix C](#).

The decontamination efficiencies obtained for each surrogate, ranging from 94.4% to 98.0%, have been used to calculate the residual concentrations of potential unknown contaminants in PET (C_{res}). By applying the decontamination efficiency percentage to the reference contamination level of 3 mg/kg PET, the C_{res} values shown in [Table 2](#) were obtained.

According to the evaluation principles (EFSA CEP Panel, 2024), the dietary exposure must not exceed 0.0025 µg/kg bw per day, below which the risk to human health is considered negligible. The C_{res} value should not exceed the modelled concentration in PET (C_{mod}) that, after 1 year at 25°C, results in a migration giving rise to a dietary exposure of 0.0025 µg/kg bw per day.

As the recycled PET is intended for the manufacturing of articles (e.g., bottles) to be used in direct contact with drinking water, the exposure scenario for infants has been applied for the calculation of C_{mod} (Exposure Scenario A; water could be used to prepare infant formula). The exposure scenario for toddlers was also applied (Exposure Scenario B; for all food contact applications except drinking water). A maximum dietary exposure of 0.0025 µg/kg bw/day corresponds to a maximum migration of 0.0481 µg/kg ($= 5 \times 0.00962$ µg/kg) or 0.0962 µg/kg ($= 10 \times 0.00962$ µg/kg) (depending on the molar mass of the contaminant substance¹⁶) into infant's food and a maximum migration of 0.156 µg/kg ($= 5 \times 0.0312$ µg/kg) or 0.312 µg/kg ($= 10 \times 0.0312$ µg/kg) (depending on the molar mass of a contaminant substance) into toddler's food. These parameters have been used to calculate C_{mod} (EFSA CEP Panel, 2024). C_{res} reported in [Tables 2](#) and [3](#) are calculated for 55% recycled PET (scenario for infants) and for 100% recycled PET (scenario for toddlers), for which the risk to human health is demonstrated to be negligible. The relationship between the key parameters for the evaluation scheme is reported in [Appendix B](#).

TABLE 2 Calculations for the infant scenario: Decontamination efficiencies from the challenge test, residual concentrations of the surrogates (C_{res}) related to the reference contamination level and calculated concentrations of the surrogates in PET (C_{mod}) corresponding to a modelled migration of 0.0481 µg/kg or 0.0962 µg/kg after 1 year at 25°C (C_{mod}).

Surrogates	Decontamination efficiency (%)	C_{res} for 55% rPET (mg/kg PET)	C_{mod} (mg/kg PET) scenario A
Toluene	97.6	0.04	0.04
Chlorobenzene	98.0	0.03	0.05
Methyl salicylate	98.0	0.03	0.12
Phenylcyclohexane	95.2	0.08	0.13
Benzophenone	94.4	0.09	0.15
Methyl stearate	96.5	0.06	0.29

Abbreviations: PET, poly(ethylene terephthalate); rPET, recycled poly(ethylene terephthalate).

¹⁵In accordance with Annex I, Table 1 of Regulation (EU) 2022/1616.

¹⁶Correction factors of 5 and 10 are applied for contaminants with molecular masses ≤ 150 Da and > 150 Da, respectively (EFSA CEP Panel, 2024).

TABLE 3 Calculations for the toddler scenario: Decontamination efficiency from the challenge test, residual concentrations of the surrogates (C_{res}) related to the reference contamination level and calculated concentrations of the surrogates in PET (C_{mod}) corresponding to a modelled migration of 0.156 or 0.312 $\mu\text{g/kg}$ after 1 year at 25°C (C_{mod}).

Surrogates	Decontamination efficiency (%)	C_{res} for 100% rPET (mg/kg PET)	C_{mod} (mg/kg PET) scenario B
Toluene	97.6	0.07	0.13
Chlorobenzene	98.0	0.06	0.15
Methyl salicylate	98.0	0.06	0.40
Phenylcyclohexane	95.2	0.14	0.42
Benzophenone	94.4	0.17	0.49
Methyl stearate	96.5	0.11	0.95

Abbreviations: PET, poly(ethylene terephthalate); rPET, recycled poly(ethylene terephthalate).

On the basis of the provided data from the challenge test and the applied conservative assumptions, the Panel considered that under the given operating conditions the recycling process Starlinger viscotec deCON is able to ensure that the level of migration of unknown contaminants from the recycled PET into food is below the conservatively modelled migration of:

- 0.0481 or 0.0962 $\mu\text{g/kg}$ (depending on the molar mass of a contaminant substance) into infant's food, at which the risk to human health is considered negligible when the recycled PET is used at up to 55% in mixtures with virgin PET for manufacturing bottles for drinking water, for long-term storage at room temperature or below, with or without hot-fill (scenario of infants).
- 0.156 or 0.312 $\mu\text{g/kg}$ (depending on the molar mass of a contaminant substance) into toddler's food, at which the risk to human health is considered negligible when the recycled PET is used at up to 100% for the manufacture of other materials and articles for contact with all types of foodstuffs except drinking water, for long-term storage at room temperature or below, with or without hot-fill (scenario of toddlers).

4 | CONCLUSIONS

The Panel considered that the process Starlinger viscotec deCON (configurations named as listed in [Appendix C](#)) is adequately characterised and that the main steps used to recycle the PET flakes into decontaminated PET flakes have been identified. Having examined the challenge test provided, the Panel concluded that temperature, residence time, pressure and reduced gas flow rate of steps 2 (preheating) and 3 (decontamination) are critical for the decontamination efficiency.

The Panel concluded that the process is capable of reducing contamination of post-consumer food contact PET to a concentration that does not give rise to concern for a risk to human health if:

- it is operated under conditions that are at least as severe as those applied in the challenge test used to measure the decontamination efficiency of the process;
- the input material of the process is washed and dried post-consumer PET flakes originating from materials and articles that have been manufactured in accordance with the EU legislation on food contact materials and contain no more than 5% of PET from non-food consumer applications;
- the recycled PET is used at up to 55% for the manufacture of materials and articles for contact with all types of foodstuffs, including bottles for drinking water, and is used for long-term storage at room temperature or below, with or without hot-fill;
- the recycled PET is used at up to 100% for the manufacture of materials and articles for contact with all types of foodstuffs, except drinking water, and is used for long-term storage at room temperature or below, with or without hot-fill.

The final articles made of this recycled PET are not intended to be used in microwave and conventional ovens and such uses are not covered by this evaluation.

5 | RECOMMENDATION

The Panel recommended periodic verification that the input to be recycled originates from materials and articles that have been manufactured in accordance with the EU legislation on food contact materials and that the proportion of PET from non-food consumer applications is no more than 5%. This adheres to good manufacturing practice and the Commission Regulation (EU) 2022/1616. Critical steps in recycling should be monitored and kept under control. In addition, supporting documentation should be available on how it is ensured that the critical steps are operated under conditions at least as severe as those in the challenge test used to measure the decontamination efficiency of the process.

6 | DOCUMENTATION PROVIDED TO EFSA

Dossier 'Starlinger viscotec deCON'. July 2023. Submitted on behalf of Starlinger & Co. GmbH., Austria.

Additional information, October 2024. Submitted on behalf of Starlinger & Co. GmbH., Austria.

Additional information, February 2025. Submitted on behalf of Starlinger & Co. GmbH., Austria.

Additional information, November 2025. Submitted on behalf of Starlinger & Co. GmbH., Austria.

ABBREVIATIONS

bw	body weight
CEP	Panel on Food Contact Materials, Enzymes and Processing Aids
C_{mod}	modelled concentration in PET
C_{res}	residual concentration in PET
FCM	Panel on Food Contact Materials
PA	polyamide
PC	polycarbonate
PET	poly(ethylene terephthalate)
PS	polystyrene
PVC	poly(vinyl chloride)
rPET	recycled poly(ethylene terephthalate)
SSP	solid state polycondensation

REQUESTOR

Austrian Competent Authority (Bundesministerium für Soziales, Gesundheit, Pflege und Konsumentenschutz)

QUESTION NUMBER

EFSA-Q-2023-00462

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PANEL MEMBERS

Claude Lambré, Riccardo Crebelli, Maria da Silva, Koni Grob, Evgenia Lampi, Maria Rosaria Milana, Marja Pronk, Gilles Rivière, Mario Ščetar, Georgios Theodoridis, Els Van Hoeck and Nadia Waegeneers.

WAIVER

In accordance with Article 21 of the Decision of the Executive Director on Competing Interest Management a waiver was granted to an expert of the Working Group. Pursuant to Article 21(6) of the aforementioned Decision, the concerned expert was allowed to take part in the preparation and discussion of the scientific output but was not allowed to take up the role of rapporteur within that time frame. Any competing interests are recorded in the respective minutes of the meetings of the FCM Panel Working Group on Recycling Plastics.

LEGAL NOTICE

Relevant information or parts of this scientific output have been blackened in accordance with the confidentiality requests formulated by the applicant pending a decision thereon by EFSA. The full output has been shared with the European Commission, EU Member States (if applicable) and the applicant. The blackening may be subject to review once the decision on the confidentiality requests is adopted by EFSA and in case it rejects some of the confidentiality requests.

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APPENDIX A

Specifications of the pre-processed input material* as provided by the applicant¹⁷

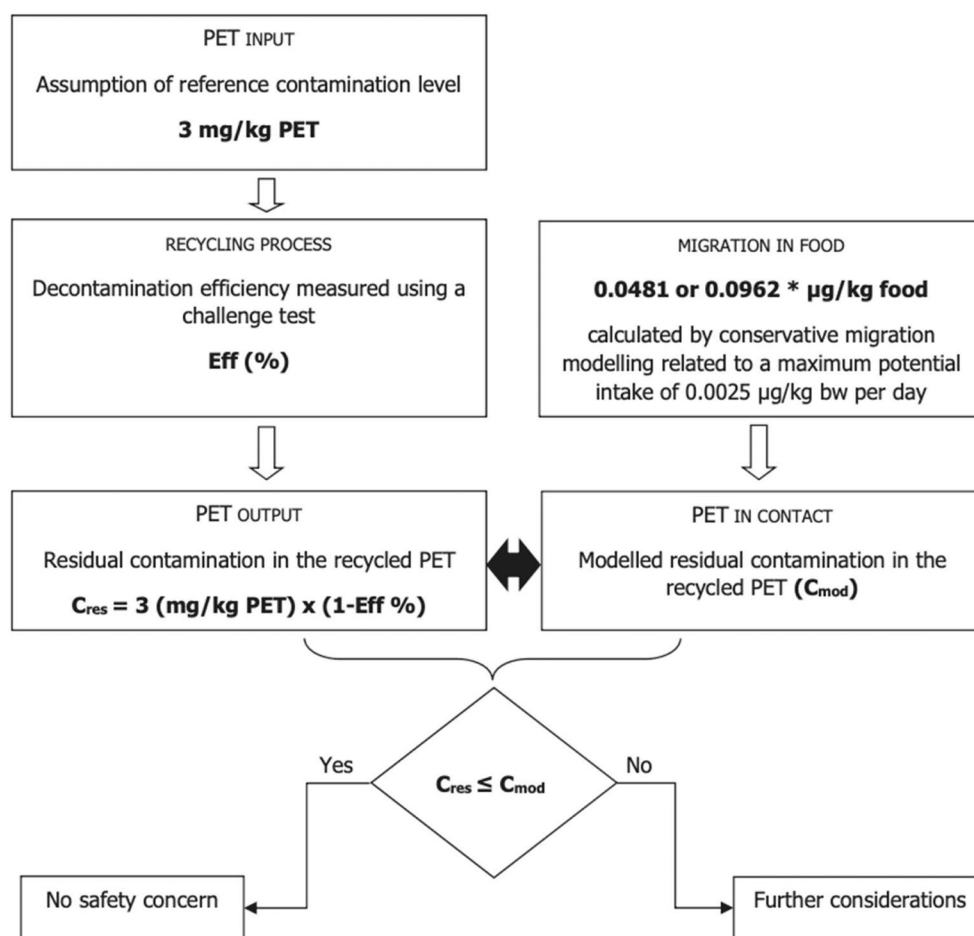
Parameter	Value
Moisture	< 1.5%
Bulk density	250–750 kg/m ³
Flake size	1–15 mm
Flake thickness	50–2000 µm
Moisture	< 1.5%
PVC	< 100 mg/kg
Polyolefins	< 200 mg/kg
PS	< 100 mg/kg
PC	< 50 mg/kg
PA	< 4000 mg/kg
Flakes with glue	< 4000 mg/kg
Wood, Paper content	< 100 mg/kg
Metal content (Aluminium, ferrous, others)	< 500 mg/kg

Abbreviations: PA, polyamide; PC, polycarbonate; PS, polystyrene; PVC, poly(vinyl chloride).
*Foreign plastics in the flakes are determined by 'oven testing'.

¹⁷Technical dossier, section 'Characterisation of the input'.

APPENDIX B

Relationship between the key parameters for the evaluation scheme, based on the most conservative scenario A (EFSA CEP Panel, 2024)



*Depending on the molecular mass of the surrogate substance. The figures are derived from the application of the human exposure threshold value of 0.0025 µg/kg bw per day applying the factors of 5 and 10 related to the overestimation of modelling (most conservative Scenario A).

APPENDIX C

Table of operational parameters¹⁸

Although the operational parameters are reported for all the process steps, the critical steps and the corresponding parameters of the challenge test/process, considered for the evaluation of the process and for which it has been concluded that the process is safe, are highlighted in green.

The process should be operated at conditions at least as severe as the ones indicated in green in the table (e.g. lower pressures are more severe than higher, higher temperatures are more severe than lower, longer times are more severe than shorter, higher gas flows generally are more severe than lower).

The official enforcement control shall verify that the recycling plant is operating in a way that complies with its authorisation. Depending on the technology, some of the parameters are inter-related and changing one parameter to a more severe value may impact another parameter into a less severe value. Therefore, eventual deviations from the values of the parameters indicated as critical (marked in green in the table) should be demonstrated not impacting significantly on the safety assessment. The table does not necessarily report all the tolerances for the operational parameters.

Process Starlinger viscotec deCON (RECYC339)								
Parameters	Step 2: Preheating to decontamination temperature				Step 3: SSP under gas flow			
	t (min)	P (mbar)	Reduced gas flow rate (m ³ /h per kg PET)	T (°C)	t (min)	P (mbar)	Reduced gas flow rate (L/h per kg PET)	T (°C)
Challenge test (Report No. 2401215/23880-2e)	■	Atmospheric	■	■	■	■ for ■ min ■ for ■ min*	≥ ■ for ■ % of time	■
	Batch				Batch			
Process	≥ ■	Atmospheric	> ■	≥ ■	≥ ■	■ % of time ≤ ■ ■ % of time ≤ ■	≥ ■ for ■ % of time	≥ ■
	Batch				Batch or semi-continuous			

* ■ % of time at ■ mbar and ■ % of time at ■ mbar.

¹⁸Technical dossier, Sections 'Table of operating Parameters' and 'Recycling process'.

Throughput and geometrical characteristics of Starlinger viscotec deCON reactors covered by this opinion

Starlinger viscotec deCON 20 configuration					
	Step	Throughput [kg/h]	Filling percentage [%]	Capacity of the reactor [m ³]	Diameter/height [m/m]
deCON20	3	Max. [redacted] (with [redacted] pre-heater)	[redacted]	[redacted]	[redacted]
Starlinger viscotec deCON 50 type of configurations					
	Step	Throughput [kg/h]	Filling percentage [%]	Capacity of the reactor [m ³]	Diameter/height [m/m]
deCON 50	3	Max. [redacted] (with [redacted] pre-heaters)	[redacted]	[redacted]	[redacted]
deCON 65	3	Max. [redacted] (with [redacted] pre-heaters)	[redacted]	[redacted]	[redacted]
deCON 50 iV+	3	Max. [redacted] (with [redacted] pre-heaters)	[redacted]	[redacted]	[redacted]
deCON 75 iV+	3	Max. [redacted] (with [redacted] pre-heaters)	[redacted]	[redacted]	[redacted]
viscoSTAR 120*	3	Max. [redacted] (with [redacted] pre-heaters)	[redacted]	[redacted]	[redacted]
viscoSTAR 150*	3	Max. [redacted] (with [redacted] pre-heaters)	[redacted]	[redacted]	[redacted]
viscoSTAR 200*	3	Max. [redacted] (with [redacted] pre-heaters)	[redacted]	[redacted]	[redacted]
viscoSTAR 250*	3	Max. [redacted] (with [redacted] pre-heaters)	[redacted]	[redacted]	[redacted]
viscoSTAR 350*	3	Max. [redacted] (with [redacted] pre-heaters)	[redacted]	[redacted]	[redacted]

*Decontamination of pellets is not covered by this assessment.